Algorithms

DYNAMIC PROGRAMMING

Last Week

- Dynamic Programming

This Week

- Dynamic Programming – Cont.
- Final Exam (in 2 weeks!)

DP Algorithm – Reminder

General structure of an algorithm based on DP:

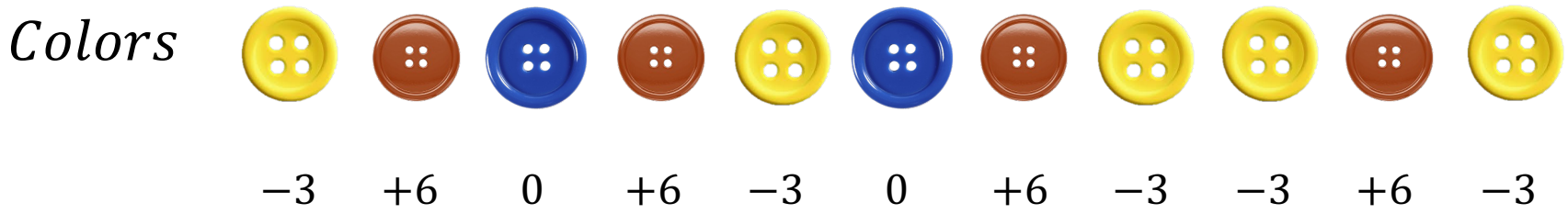
- Define the table.
- **Identify base cases** (sub-problems that can be filled directly).
- Write a **recursive formula**, and then, instead of performing the recursion, **fill the table**.
- Conclude the **value of the solution** from the table.
- Retrieve the solution.

Exercise 1

A row of n buttons is placed on the table. Every button, $1 \leq i \leq n$, has a color

$$c_i \in \{Red, Blue, Yellow\}.$$

You need to selects two indices (j, k) such that $1 \leq j \leq k \leq n$. The value of a selection is determined as follows: The set of buttons $j, \ldots, k$ are considered; every **Red button adds 6 points**, every **Yellow button reduces 3 points**, and **Blue buttons are ignored**.



Exercise 1

A row of n buttons is placed on the table. Every button, $1 \leq i \leq n$, has a color

$$c_i \in \{Red, Blue, Yellow\}.$$

You need to selects two indices (j, k) such that $1 \leq j \leq k \leq n$. The value of a selection is determined as follows: The set of buttons $j, \ldots, k$ are considered; every **Red button adds 6 points**, every **Yellow button reduces 3 points**, and **Blue buttons are ignored**.

Formally, your goal is to select indices j ? k that maximize the following expression:

$$6 \cdot |i: c_i = Red, j \leq i \leq k| - 3|i: c_i = Yellow, j \leq i \leq k|$$

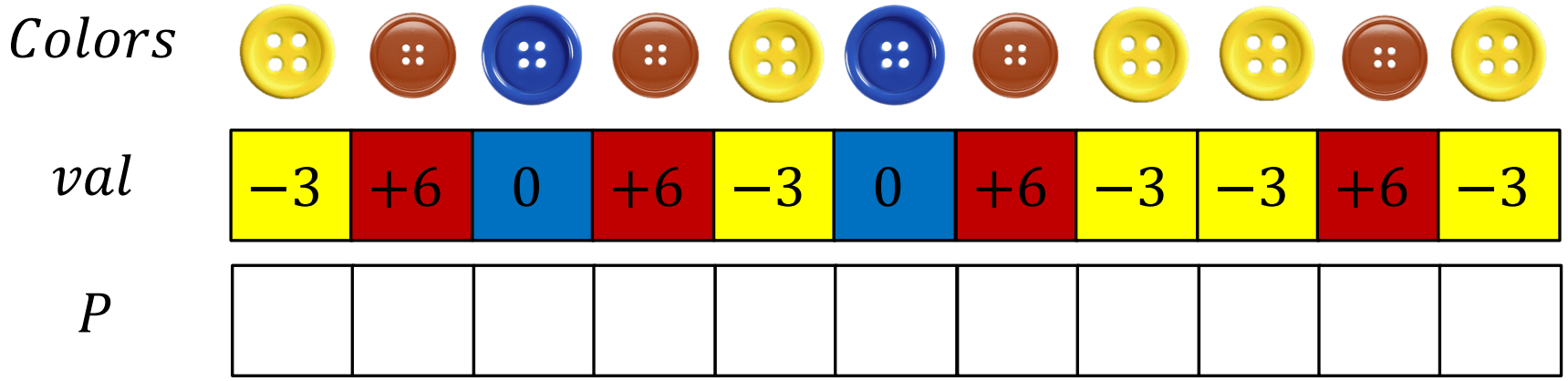
Suggest an $O(n)$-time algorithm (using dynamic programming) that calculates the maximal possible score, and finds the indices (j, k) achieving it.

Exercise 1

Let $P[i]$ be the maximal value of buttons that can be obtained from a sequence that starts with i.

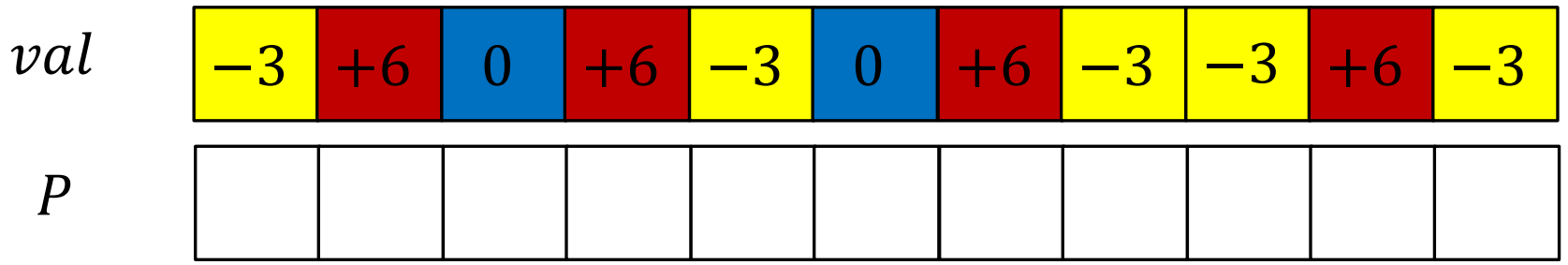
For simplicity, define a function val: $\{Red, Blue, Yellow\} \rightarrow \{+6, 0, -3\}$ with

$$val(Red) = 6,\ val(Blue) = 0,\ val(Yellow) = -3.$$



Exercise 1 - Solution

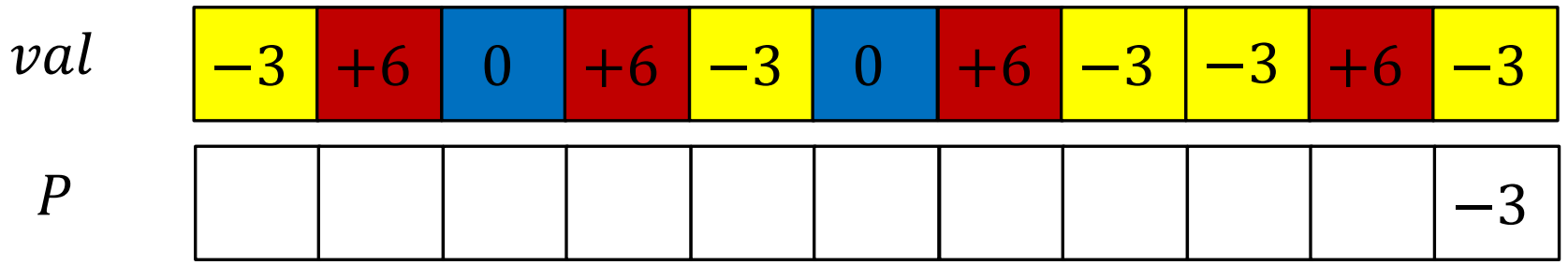
Base case:



Exercise 1 - Solution

Base case:

$$P[n] = val[c_n]$$

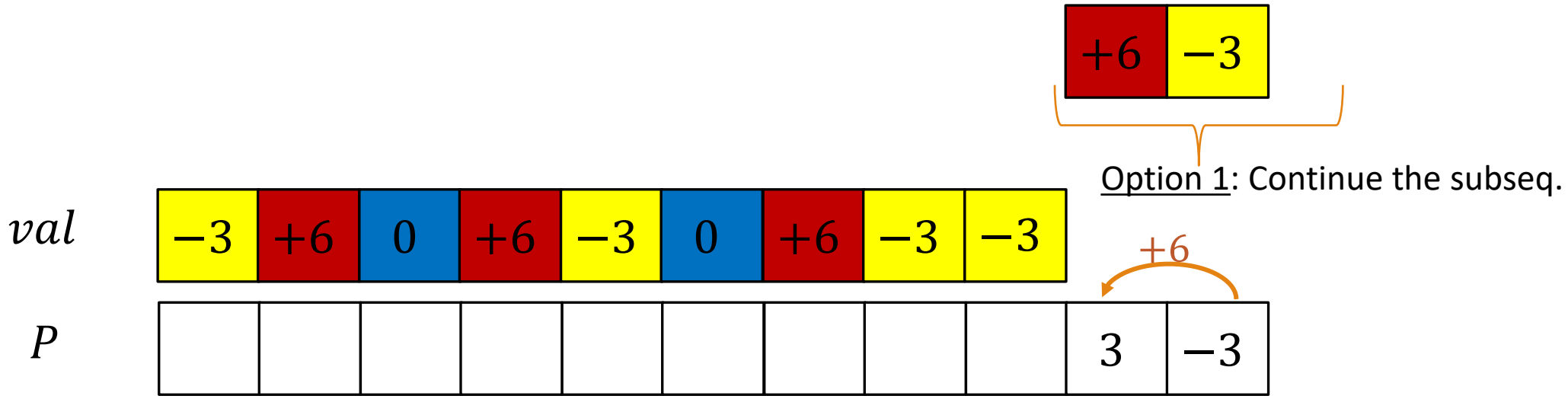


Exercise 1 - Solution

Base case:

$$P[n] = val[c_n]$$

Recursive formula:

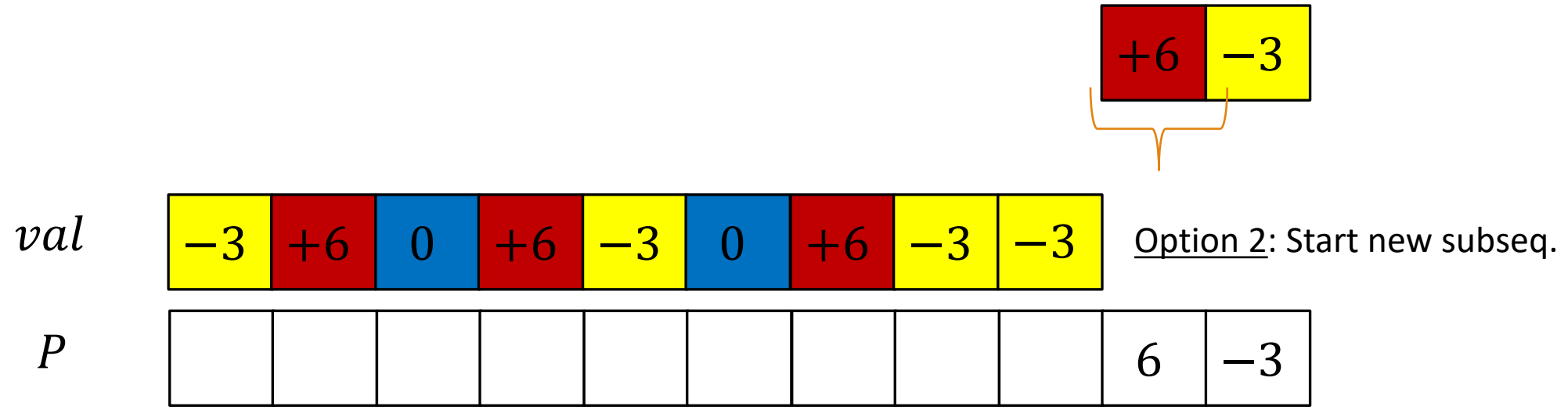


Exercise 1 - Solution

Base case:

$$P[n] = val[c_n]$$

Recursive formula:



Option 2: Start new subseq.

Exercise 1 - Solution

Base case:

$$P[n] = val[c_n]$$

Recursive formula:

$$For\ 1 \leq i < n,\ P[i] = \max\{val(c_i), val(c_i) + P[i+1]\}$$

val	-3	$+6$	0	$+6$	-3	0	$+6$	-3	-3	$+6$	-3
P	12	15	9	9	3	6	6	0	3	6	-3

Exercise 1 - Solution

Base case:

$$P[n] = val[c_n]$$

Recursive formula:

$$For\ 1 \leq i < n,\ P[i] = \max\{val(c_i), val(c_i) + P[i+1]\}$$

Solution Value: $\max\{P[i]\}$

val	-3	$+6$	0	$+6$	-3	0	$+6$	-3	-3	$+6$	-3
P	12	15	9	9	3	6	6	0	3	6	-3

Exercise 1 - Solution

Algorithm:

1. Create an array of length n and fill it according to the recursive formula, in the correct order.

2. Finding the solution Value:
   - Scan the table P, the entry for which $P[i]$ is maximal is 'j' (the first element in the winning sub-sequence).
   - The maximal value is $P[j]$.

3. Finding a solution:
   - Scan the table P starting at 'j'. Continue as long as $P[i] = val(c_i) + P[i+1]$.
   - Once you reach an entry i for which $P[i] = val(c_i)$, set 'k'$= i$ (the last element in the winning sub-sequence).

Exercise 1 - Solution

Complexity:

The table is an array of length n.

The base case fills the first entry, and the algorithm proceeds by filling the table sequentially (backwards), each entry in $O(1)$.

Finding the indices takes $O(n)$.

All together $O(n)$.

Exercise 2 - Intro

What is Palindrome?

A palindrome is a sequence of characters, numbers, or symbols that reads the same forwards and backwards.

In the context of strings, if you reverse the string S, the result is identical to the original string:

$$S = reverse(S)$$

Exercise 2

Given a string A of length n, find the ***length*** of the longest palindromic subsequence in it.

Example: If the given string is “BBCBABACB”, then the output should be 7 as “BCBABCB” is the longest palindromic subsequence in it. “BBBB” and “BBABB” are also palindromic subsequences of the given sequence, but not the longest ones.

BBCBABACB

BBCBABACB

BBCBABACB

Exercise 2 - Solution

- Table: Let $B[i,j]$ be the longest palindromic subsequence of the sub-word $a_i, \ldots, a_j$.

BBCBABACB

$B[2,7]$ = length of longest palindromic subsequence in BCBABA

$B[2,7] = 3$

Exercise 2 - Solution

- Table: Let $B[i, j]$ be the longest palindromic subsequence of the sub-word $a_i, \ldots, a_j$.
- Base case: $B[i, i] = 1$, $B[j, i] = 0$ $(j < i)$

Exercise 2 - Solution

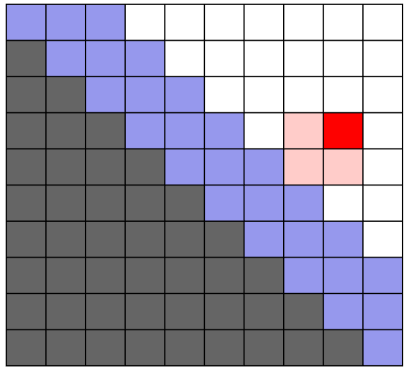
Consider the following string: “BBCBABACB”

Initialize the table:

i/j	1	2	3	4	5	6	7	8	9
1	1								
2	0	1							
3	0	0	1						
4	0	0	0	1					
5	0	0	0	0	1				
6	0	0	0	0	0	1			
7	0	0	0	0	0	0	1		
8	0	0	0	0	0	0	0	1	
9	0	0	0	0	0	0	0	0	1

Exercise 2 - Solution

- Table: Let $B[i,j]$ be the longest palindromic subsequence of the sub-word $a_i, \dots, a_j$.
- Base case: $B[i,i] = 1$, $B[j,i] = 0$ $(j < i)$
- For $i, j = 1, \dots, n$:



$$B[i,j] = \begin{cases} \max\{B[i+1,j], B[i,j-1]\} & j > i \text{ and } a_i \neq a_j \\ \max\{B[i+1,j], B[i,j-1], B[i+1,j-1]+2\} & j > i \text{ and } a_i = a_j \end{cases}$$

- Return $B[1,n]$.

Exercise 2 - Solution

Fill first diagonal

$$B[i,j] = \begin{cases} \max\{B[i+1,j], B[i,j-1]\} & a_i \neq a_j \\ \max\{B[i+1,j], B[i,j-1], B[i+1,j-1]+2\} & a_i = a_j \end{cases}$$

$$B[1,2] = \max\{B[2,2], B[1,1], B[2,1\,]+2\} = \max\{1,1,0+2\} = 2$$

$$B[2,3] = \max\{B[3,3], B[2,2]\} = 1$$

"BBCBABACB"

i/j	1	2	3	4	5	6	7	8	9
1	1	2							
2	0	1	1						
3	0	0	1	1					
4	0	0	0	1	1				
5	0	0	0	0	1	1			
6	0	0	0	0	0	1	1		
7	0	0	0	0	0	0	1	1	
8	0	0	0	0	0	0	0	1	1
9	0	0	0	0	0	0	0	0	1

Exercise 2 - Solution

Fill second diagonal

$$B[i,j] = \begin{cases} \max\{B[i+1,j], B[i,j-1]\} & a_i \neq a_j \\ \max\{B[i+1,j], B[i,j-1], B[i+1,j-1]+2\} & a_i = a_j \end{cases}$$

$B[1,3] = \max\{B[2,3], B[1,2]\} = 2$

$B[2,4] = \max\{B[2,3], B[3,4], B[3,3]+2\} = 3$

$B[3,5] = \max\{B[4,5], B[3,4]\} = 1$

"BBCBABACB"

i/j	1	2	3	4	5	6	7	8	9
1	1	2	2						
2	0	1	1	3					
3	0	0	1	1	1				
4	0	0	0	1	1	3			
5	0	0	0	0	1	1	3		
6	0	0	0	0	0	1	1	1	
7	0	0	0	0	0	0	1	1	1
8	0	0	0	0	0	0	0	1	1
9	0	0	0	0	0	0	0	0	1

Exercise 2 - Solution

Fill the rest and return $B[1,9]$

$$B[i,j] = \begin{cases} \max\{B[i+1,j], B[i,j-1]\} & a_i \neq a_j \\ \max\{B[i+1,j], B[i,j-1], B[i+1,j-1]+2\} & a_i = a_j \end{cases}$$

"BBCBABACB"

i/j	1	2	3	4	5	6	7	8	9
1	1	2	2	3	3	5	5	5	7
2	0	1	1	3	3	3	3	5	7
3	0	0	1	1	1	3	3	5	5
4	0	0	0	1	1	3	3	3	5
5	0	0	0	0	1	1	3	3	3
6	0	0	0	0	0	1	1	1	3
7	0	0	0	0	0	0	1	1	1
8	0	0	0	0	0	0	0	1	1
9	0	0	0	0	0	0	0	0	1

Exercise 2 - Solution

Complexity:

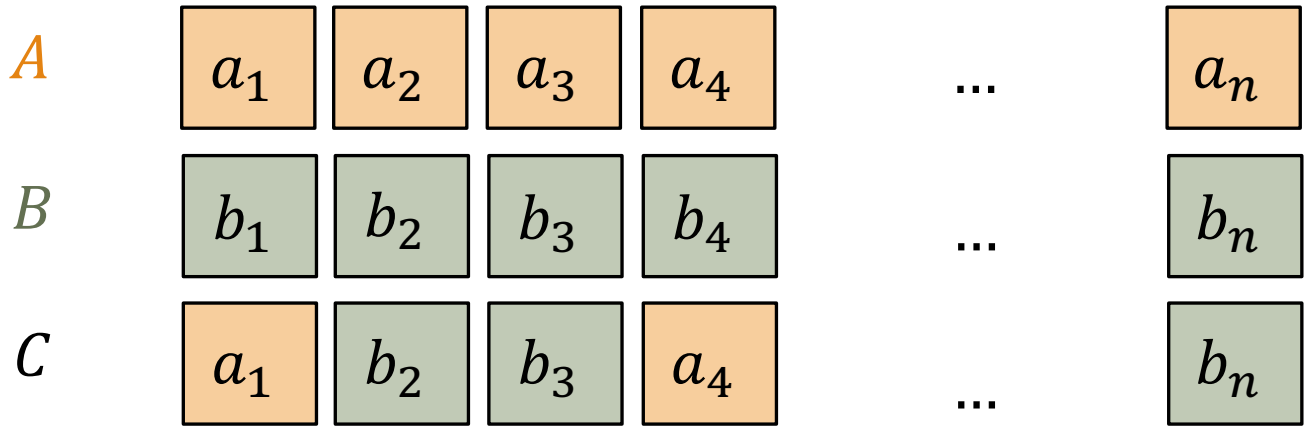
- Need to fill $O(n^2)$ cells in the table.
- Each cell takes $O(1)$ (checking if $a_i = a_j$ and finding the maximum).
- Total running time: $O(n^2)$.

Exercise 3

Given two sequences $A = a_1, a_2, \dots, a_n$ and $B = b_1, b_2, \dots, b_n$ we say that a sequence C is a combination of A, B if and only if for every $1 \leq i \leq n$ we have

$$c_i = a_i \text{ or } c_i = b_i$$

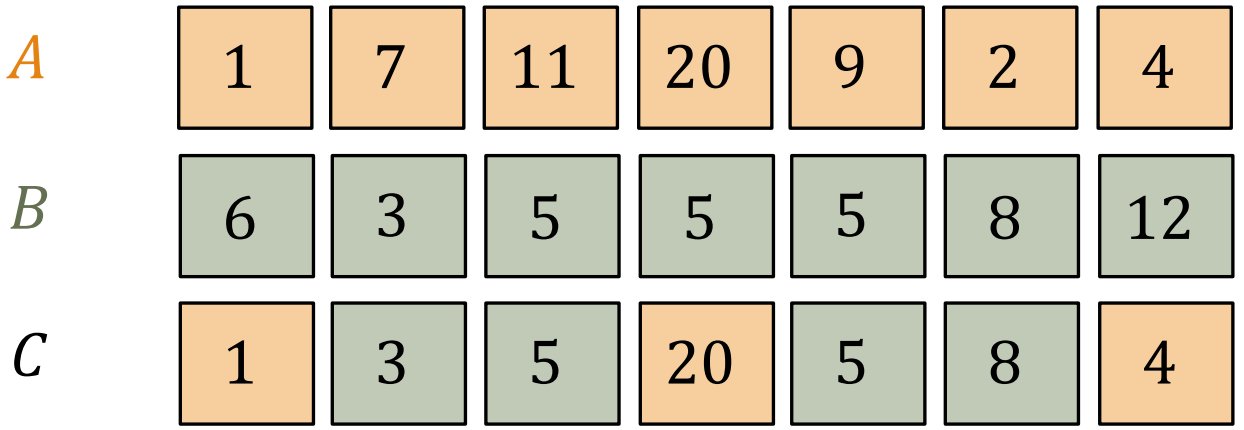
In other words, for i from 1 to n, we pick either a_i or b_i to be c_i.



Exercise 3

Denote the total sum of elements in a sequence S by $SUM(S)$.

Given two sequences A, B of n natural numbers, and given a target sum natural s, suggest an algorithm for determining whether there exists a combination C of A and B such that $SUM(C) = s$.



$s = 46$

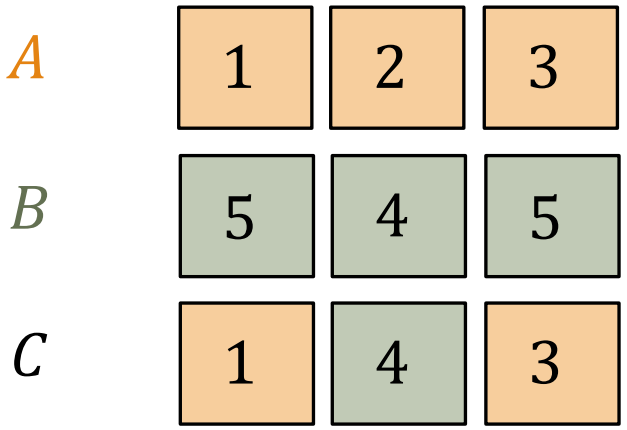
$SUM(C) = 46$
$\Rightarrow return\ TRUE$

Directions: Use Dynamic Programming with a Boolean table H of size $n \times s$.

Exercise 3

Denote the total sum of elements in a sequence S by $SUM(S)$.

Given two sequences A, B of n natural numbers, and given a target sum natural s, suggest an algorithm for determining whether there exists a combination C of A and B such that $SUM(C) = s$.



$s = 8$

$SUM(C) = 8$
$\Rightarrow return\ TRUE$

$s = 7$

$\forall C, SUM(C) \neq 7$
$\Rightarrow return\ FALSE$

Directions: Use Dynamic Programming with a Boolean table H of size $n \times s$.

Exercise 3

The Table:

$$H[i,k] = \begin{cases} TRUE & \exists C_i = (c_1, c_2, \dots, c_i)\ s.t. \forall 1 \le j \le i, c_j = a_j \vee b_j\ and\ SUM(C_i) = k \\ FALSE & o.w. \end{cases}$$

Characters \ Sum	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	…	$k=s$
$i=0$									
$i=1$									
$i=2$									
…									
$i=n$									

Exercise 3

The Table:

$$H[i,k] = \begin{cases} TRUE & \exists C_i = (c_1, c_2, \dots, c_i)\ s.t. \forall 1 \le j \le i, c_j = a_j \vee b_j\ and\ SUM(C_i) = k \\ FALSE & o.w. \end{cases}$$

Characters \ Sum	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	…	$k=s$
$i=0$									
$i=1$									
$i=2$									
…									
$i=n$									

Base Cases?

Exercise 3 - Solution

Base cases:

$H[0,0] = true$

Characters \ Sum	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	…	$k = s$
$i = 0$									
$i = 1$									
$i = 2$									
…									
$i = n$									

Exercise 3 - Solution

Base cases:

$H[0,0] = true$

Characters \ Sum	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	…	$k = s$
$i = 0$	1								
$i = 1$									
$i = 2$									
…									
$i = n$									

Exercise 3 - Solution

Base cases:

$H[0,0] = true$

$H[0,k] = false,\ since\ k \neq 0$ (s is natural)

Characters \ Sum	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	…	$k=s$
$i=0$	1	0	0	0	0	0	0	…	0
$i=1$									
$i=2$									
…									
$i=n$									

Exercise 3 - Solution

Base cases:

$H[0,0] = true$

$H[0,k] = false,\ since\ k \neq 0$ (s is natural)

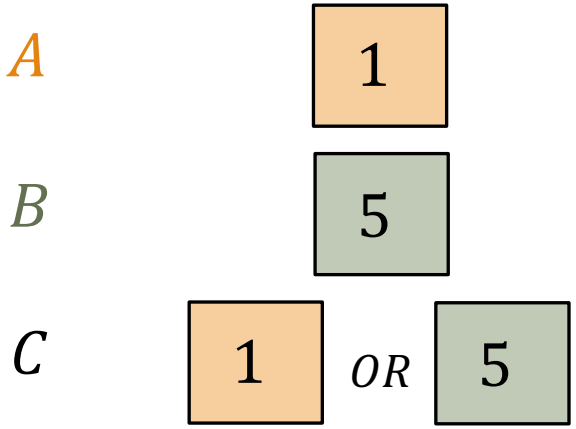
$H[i,0] = false,\ for\ i \neq 0$ (all numbers are naturals)

Characters \ Sum	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	…	$k=s$
$i=0$	1	0	0	0	0	0	0	…	0
$i=1$	0								
$i=2$	0								
…	…								
$i=n$	0								

Exercise 3 - Solution

The recursive formula:

For the first row, a_1 or b_1 can be selected to be c_1



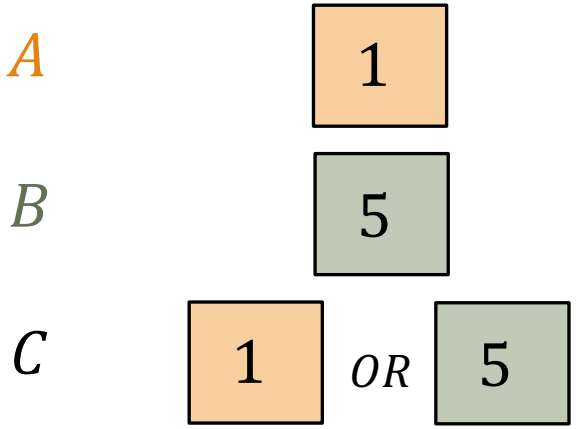
$$SUM(C_1) = SUM(C_0) + (a_1\ OR\ b_1)$$

$$H[1,1] = TRUE$$
$$H[1,5] = TRUE$$
$$H[1, k \neq (1 \vee 5)] = FALSE$$

Exercise 3 - Solution

The recursive formula:

For the first row, a_1 or b_1 can be selected to be c_1



$$SUM(C_1) = SUM(C_0) + (a_1\ OR\ b_1)$$

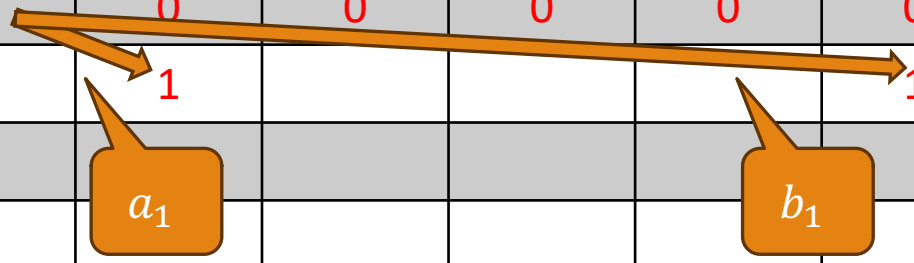
$$H[1,1] = H[0,0]$$
$$H[1,5] = H[0,0]$$
$$H[1, k \neq (1 \vee 5)] = H[0, k \neq 0]$$

Exercise 3 - Solution

The recursive formula:

For the first row, a_1 or b_1 can be selected to be c_1

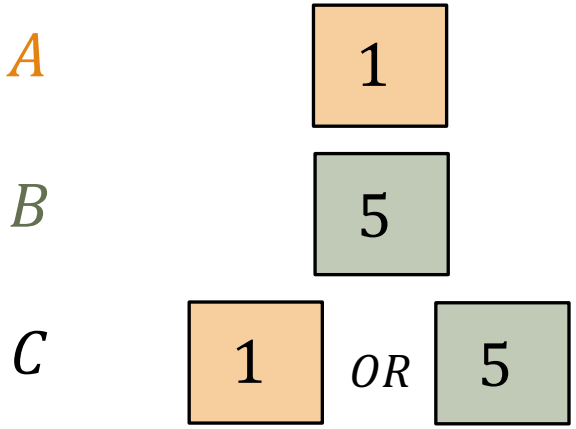
Characters \ Sum	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	…	$k=s$
$i=0$	1	0	0	0	0	0	0	…	0
$i=1$	0	1				1			
$i=2$	0								
…	…								
$i=n$	0								



Exercise 3 - Solution

The recursive formula:

For the first row, a_1 or b_1 can be selected to be c_1



$$SUM(C_1) = SUM(C_0) + (a_1\ OR\ b_1)$$

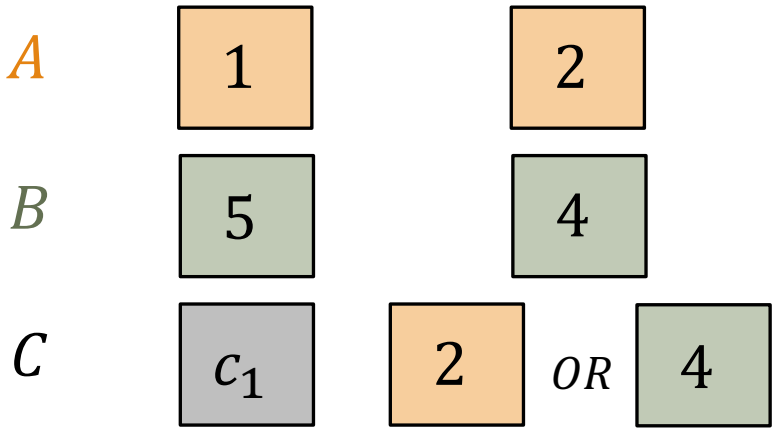
$$H[1,1] = H[0,0]$$
$$H[1,5] = H[0,0]$$
$$H[1, k \neq (1 \vee 5)] = H[0, k \neq 0]$$

$$\Rightarrow H[1,k] = \mathrm{H}[0, \mathrm{k} - \mathrm{a}_1] \vee \mathrm{H}[0, \mathrm{k} - \mathrm{b}_1]$$

Exercise 3 - Solution

The recursive formula:

For the second row, a_2 or b_2 can be selected to be c_2



$$SUM(C_2) = SUM(C_1) + (a_2\ OR\ b_2)$$

$$k = 3 \rightarrow H[2,3] = TRUE$$
Because $H[2,3] = H[1,1]$

$$k = 5 \rightarrow H[2,5] = TRUE$$
Because $H[2,5] = H[1,1]$

$$k = 6 \rightarrow H[2,6] = FALSE$$
Because $H[2,6] = H[1,4] \vee H[1,2]$

$$\Rightarrow H[2,k] = H[1, k - a_i] \vee H[1, k - b_i]$$

Exercise 3 - Solution

The recursive formula:

In the general case, a_i or b_i can be selected to be c_i

$$\mathbf{H}[\mathbf{i}, \boldsymbol{k}] = \mathbf{H}[\boldsymbol{i-1}, \boldsymbol{k-a_i}] \vee \mathbf{H}[\boldsymbol{i-1}, \boldsymbol{k-b_i}]$$

Fill the table in order (row by row, column by column).

Exercise 3 - Solution

The recursive formula:

In the general case, a_i or b_i can be selected to be c_i

$$\mathbf{H}[\mathbf{i}, \boldsymbol{k}] = \mathbf{H}[\boldsymbol{i} - \mathbf{1}, \boldsymbol{k} - \boldsymbol{a_i}] \vee \mathbf{H}[\boldsymbol{i} - \mathbf{1}, \boldsymbol{k} - \boldsymbol{b_i}]$$

Fill the table in order (row by row, column by column).

The Solution Value:

$$\mathbf{H}[\mathbf{n}, \boldsymbol{s}]$$

Exercise 3 - Solution

Complexity:

- Need to fill $O(nS)$ cells in the table.
- Each cell takes $O(1)$.
- Total running time: $O(nS)$.

Exam - What to expect?

- 6 questions:
  - MST (16-20)
  - Shortest paths (16-20) – suggest an algorithm
  - Flow (16-20)
  - DP (16-20)
  - Greedy (14-16)
  - Pre–midterm: BFS,DFS, topo, Euler, Hamilton (14-16)
- No De-Bruijn
- For MST and flow:
  - Variants of ‘Run the algo’
  - Know the theory!! One(+) technical, one(+) prove/disprove

Exam – My Two Cents

- Read the entire question before you start answering
- At home, **write** full answers.
  - Compare with the formal answer and find what's missing in yours
  - Rewrite your answer – if it helps the changes sink in.
- Practice full exams with a stopwatch
- Find a strategy that suits you
- Logistics:
  - We still have more office hours (time and place, TBD)
  - USE THE PIAZZA!!!


GOOD LUCK